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Distributed Ledger Technology-based land transaction system with trusted nodes consensus mechanism

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ABSTRACT

Blockchain technology is an emerging solution to various financial applications that require a secure and tamper-proof transactions system. One such application is the land registry. Management of transactions for land registration is a tedious task. It is highly insecure and subject to forgery land records, Issues in verification, mediators, etc. This article presents a scalable and novel property/land registration framework based on blockchain technology. The transparent and immutable nature of blockchain technology makes it reliable for intrusive and fraudulent land registration activities. The framework uses the InterPlanetary File System (IPFS), a Peer-to-Peer (P2P) swarm network. It allows the sharing and storing of data in an efficient, decentralized, and transparent manner. This article also proposes a Trusted Node Consensus Algorithm (TNCA) based on the miners' trust value that minimizes the computational overhead of broadcasting the new block. The proposed approach - trusted node consensus algorithm requires lesser time to add the block into the blockchain, and message exchange per consensus is 58.94% lesser than the traditional Proof-of-Work (PoW) approach and 26.44% lesser time than the existing load-balanced method.

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1. Introduction

Technology innovation has introduced incredible changes in commercial, industrial, academic, administration, etc. It has been able to prevent fraud and leakage in the tax collection system. One such sector that generates large revenue for the government is the land registry in the real estate sector. Whenever any land/property is transferred from one entity to another entity, changing ownership is required, called the land registration process. This process requires several steps like stamp duty, conveyance deed, sale deed, and sale agreement (Singh and Vardhan, 2019). After payment of stamp duty and taxes, buyers have to submit the property documents for ownership change. The deed must be executed at the registrar's office. Once all the property documents and some personal documents (Id proof, Address proof, etc.) are reviewed and found to be perfect, the land gets registered as a final step.

So, one has to go through all these tiresome processes for transferring property. The entire procedure usually takes a long time to process land-related and personal documents. Stamp duty and registration charges vary from state to state in the country. The state is responsible for creating policy for the land registration process, and the sub-region specifies authorities to maintain or implement these policies. The property transaction records are maintained in a centralized manner at the state level. The process of land registration requires manual and physical steps, and many authorities involvement at various levels. The land registration process has various issues such as time consumption by the whole process, excessive paperwork, incorrect data due to manual documentation process, additional cost for brokers, and fraud cases like ambiguous land titles, fake/forgery land records have also been detected that dupe the buyer. Hence, the land registration process needs to be digitalized and requires a secure, transparent, decentralized solution. Hence, Distributed Ledger Technology (DLT) based blockchain solution is seeking the attention of researchers for the implementation of various financial (Singh and Vardhan, 2019) and e-governance (Singh and Vardhan, 2019) applications. In order to address these issues, we have proposed a DLT based architecture and an efficient trust value-based consensus mechanism to verify property transactions.

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1.1. DLT and blockchain

DLT is rapidly growing as a solution for many financial applications requiring a reliable and tamper-proof transaction system. The decentralization nature of this technology leads to higher levels of fault tolerance. The blockchain-based DLT is used for storing the transactions in the form of blocks. Once any record is stored in the blockchain, it becomes immutable. And this intrinsic nature of blockchain makes itself convenient for use by many functionalities of government, as discussed by Singh et al. (Singh et al., 2018) (Singh and Vardhan, 2019). For providing security, blockchain technology uses a digital signature and cryptographic hash to generate and link blocks (Zheng et al., 2017). Before a block is added to the blockchain, it has to be verified by the authorized members of the network to prove the trustworthiness of the transaction/ information being stored into the blocks. This is where distributed consensus protocols came into the picture. According to the consensus votes, the decide authorized members (miners) whether the transaction is found to be valid or not. Only after achieving the consensus successfully can the block be added to the existing blockchain by using the preceding block hash (Vujičić et al., 2018). The next section describes the literature reviews related to the proposed work, consensus mechanisms, and applications based on DLT.

2. Related work

Blockchain technology came into the picture when Nakamoto, in 2008, introduced Bitcoin (Nakamoto, 2009). Blockchain, as the key technology behind bitcoin, is essentially a distributed database system. Zibin et al. (Zheng et al., 2017, Tschorsch and Scheuermann, 2016) explore the design, consensus, and likely trends of the blockchain. They find that blockchain works as an immutable ledger that permits the transaction to occur in a decentralized manner. Blockchain-based applications are growing rapidly, covering numerous areas, including financial services, reputation systems, and the Internet of Things (IoT), among other issues. Author's present a comprehensive overview of blockchain technology and compare some typical consensus algorithms such as Proof-of-Work (PoW), Proof-of-Stake (PoS), Practical Byzantine Fault Tolerance (PBFT), and Delegated Proof of Stake (DPoS) used in blockchains. Furthermore, technical challenges and recent advances are briefly listed.

Vujičić et al. (Vujičić et al., 2018) focus on applications of blockchain in cryptocurrencies like Bitcoin. Singh and Vardhan (Singh and Vardhan, 2019) have proposed a consensus mechanism for blockchain-based Distributed Ledger Technology (DLT) for land registration. They focus on providing a framework for reliable and decentralized P2P infrastructure for managing the e-stamp and property registration process. The authors extended their research in this area of DLT and proposed a distributed and decentralized architecture for real state transactions and verification requests (Singh and Vardhan, 2019). They propose a consensus mechanism based on majority voting. The proposed decentralized architecture provides protection against fraudulent activities in the real state world. Singh and Vardhan (Singh et al., 2019) proposed DLT based banking transactions for an e-cheque clearing system based on the DLT. The proposed approach expands the range of e-cheque from local to global banking. The proposed approach is based on a permission blockchain. All banks willing to participate in this system must join the framework to provide their customers with a faster cheque clearing facility. It is deliberately designed to allow any bank to see a cheque issued by its customers or deposited by any other bank.

Singh et al. (Singh et al., 2018) provide architecture as a solution for integrating the e-wallets of different banks using blockchain technology. In this proposed work, they use IPFS (Benet, 2014), a P2P swarm network, to implement the e-wallet system. This solu-

tion reduces the server load. Kumar et al. (Kumar et al., 2020) proposed a block validation and consensus algorithm, which randomly generates a block and adds it to the existing blockchain. This research provides architecture based on the blockchain using a one-way hash function to store and access traffic-related information on the vehicle's blockchain network. Singh and Vardhan (Singh and Vardhan, 2018, Singh and Vardhan, 2020) have proposed a DLT based e-stamp procurement system that discusses its efficiency in terms of the authenticity of the stamp paper being used for real estate transactions. Kumari and Kushwaha (Kumari and Kushwaha, 2011) have discussed the issues related to authentication and authorization of users in their paper and proposed an approach using the Collaborative Trust Enhanced Security (CTES) model for Kerberos-style authentication. Authors have extended the research in the same fields and provided a model using Kerberos by enhancing the distributed system's trustworthiness (Kumari et al., 2010). They have also proposed a scheduling algorithm for migrated jobs in trusted distributed systems (Mishra et al., 2010). Pippal et al. (Pippal et al., 2011) proposed a scalable authentication approach based on the CTES model. Scalability can be improved by load balancing in a distributed system. Mishra et al. (Mishra et al., 2009) have proposed a trusted and comprehensive hybrid load balancing mechanism for the distributed environment. IPFS that can be used to create a block is a P2P network, and various protocols have been proposed for storing and sharing data in a distributed manner. Mallick and Kushwaha (Mallick and Kushwaha, 2012) have proposed an improved version of PUSH named Ex-PUSH protocol. The proposed publication-subscribe system can deliver the notification with content delivery capabilities to the subscriber.

This section presents the usage of blockchain in the different application domains and consensus mechanisms. After the study of these researches, it is found that very few researchers exist that propose a blockchain-based P2P architecture for a property transaction system that is efficient, fraud-free, tamper-proof, and trustworthy. In existing consensus mechanisms like PoW, most of the time is consumed in solving the puzzles for authorization. For any application based on blockchain, if all network nodes participate in the consensus process, lots of CPU time and network bandwidth gets consumed and throughput will reduce drastically. Hence, there is a requirement for efficient consensus mechanisms that miner to achieve consensus. The next section describes the details of the proposed property transaction system. We have proposed a framework that deals with a land transaction management system using blockchain and an efficient consensus mechanism. The focus is on the illustration of the land transaction system.

3. Proposed DLT based property transaction system

This work proposes a scalable and novel property/land transaction framework and an efficient trust value-based consensus mechanism to provide a fast, secure, transparent, and tamperproof land registration system. The proposed architecture is based on blockchain technology. Here we have used the InterPlanetary File System (IPFS) is a distributed and Peer-to-Peer network, to implement the proposed application. Therefore all registry offices in the state willing to deploy the proposed system must join the proposed blockchain-based framework to provide a fast and transparent verification and registration process.

3.1. Proposed swarm network architecture for property transaction system

The proposed blockchain-based P2P swarm architecture shown in Fig. 3.1 is built over the IPFS. Various platforms are available to

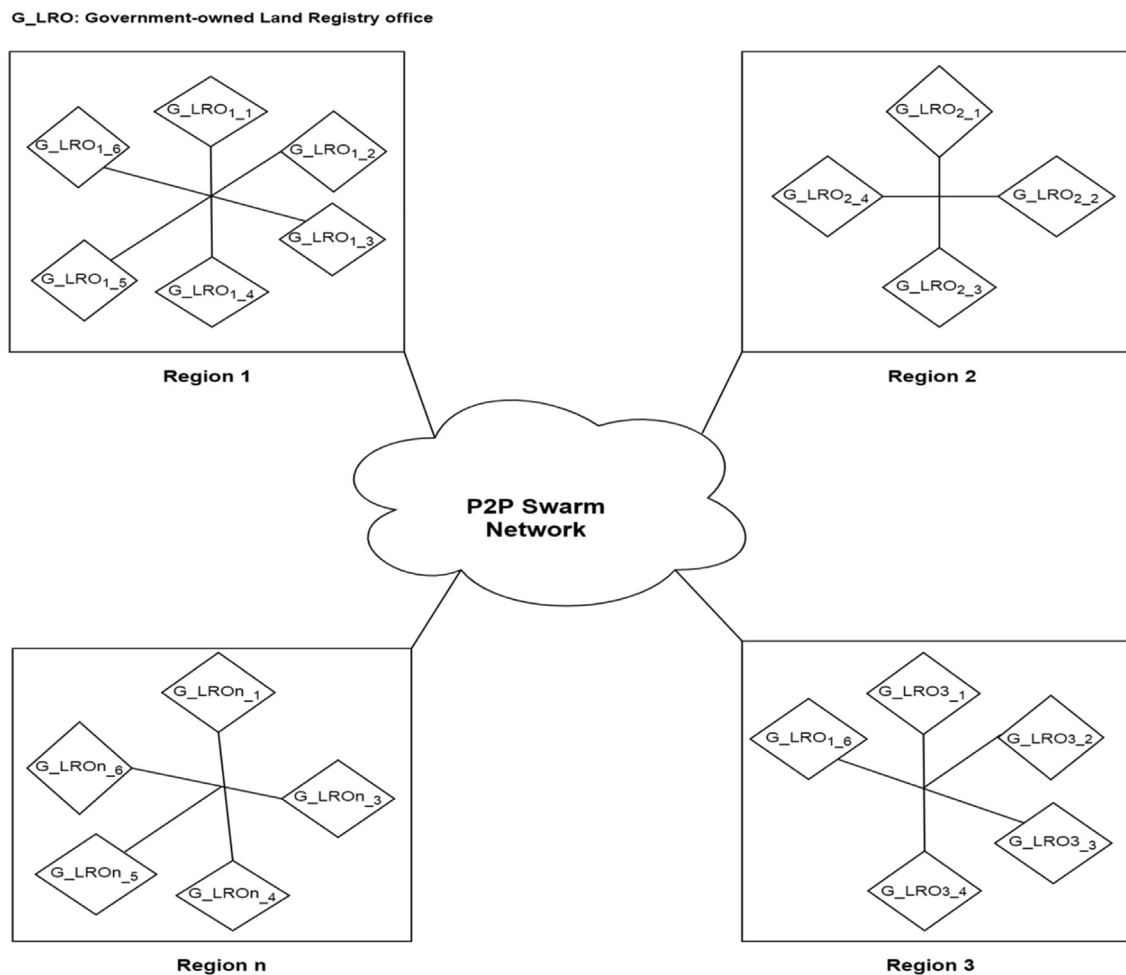


Fig. 3.1. Swarm based P2P Architecture.

implement blockchain applications, i.e., Ethereum, Coinbase, Hyperledger fabric, etc. These platforms have their own mining and security policies. IPFS provides flexibility for the developers to develop their mining policies and consensus mechanisms for the blockchain application. IPFS platform has much functionality, such as content-addressable data sharing, P2P network, and decentralization.

3.1.1. Participating entities in the proposed architecture

Participating entities are used to create, verify, and secure the land/property's transactions. These are briefly described as follows:

A. Government-owned Land Registry Office: The land transaction needs a multilevel verification process by the Government-owned Land Registry Office (G_LRO), divided into sub-registrar and registrar offices as shown in Fig. 3.2. First Sub-Registrar Office (SRO) verifies all the submitted documents, including map plans of land, ID proofs, and two witnesses. Once document verification is done, then SRO sends the verified document to the registrar's office. Registrar verifies the document and examines the formal requirements like identifying the signatures of the parties and two witnesses, correctness of the deed format, etc. The registrar's office verifies that land, buyer/seller, witnesses, and finally issues the buyer's registered deed.

B. Region: The generation of e-registry documents is the main responsibility of the regional office in regions. The region comprises of various registry offices and nodes connected through a

web server. One region corresponds to the state or union territory. The registry office has various registry office miners and authorized professional miner nodes for mining the transaction blocks.

C. Web Server: A web server is attached to a region to store the regional database of land record. The web server is specific to a region that serves as an information repository for the various registry offices corresponding to the region's district.

D. Registry Office Miner: Registry office miners are those miners who are part of the G_LRO. They are classified into two categories of Highly Trusted Registry Office Miner (HTROM) and Low Trusted Registry Office Miner (LTROM) Miner.

E. Authorized Professional Miner: The Authorized Professional Miner (APM) nodes perform a significant part of mining and transaction verification. APM provide their services to enhance the registry office's performance and scalability because the APM miner deploys high computational power. It reduces the registry office miner overhead because the regional miner is governed by the government that is busy with the deed's registration and other day to day activities.

F. Bootstrap Server: The bootstrap server provides the service to a new peer who intends to join a P2P network to discover contact information about other peers in the existing network. The bootstrap server maintains the list of the active and inactive node that is part of the network.

The proposed land registry framework is assumed to be entirely possessed by the government, which controls all the system entities. Numerous registry offices and their web servers form the

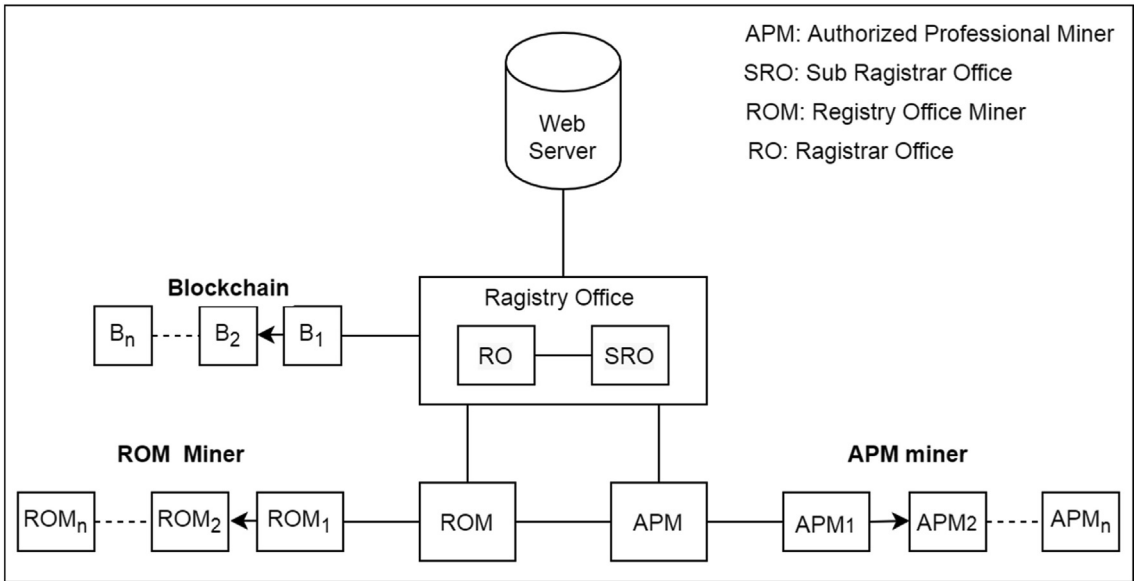


Fig. 3.2. Government-owned Land Registry Office of a Region/State (G_LRO).

backbone of the proposed P2P swarm architecture. A web server exists in each registry office, which also has a copy of the updated blockchain. Besides, some professional miners, who utilize state-of-the-art technologies, are also involved in the transaction process. In the proposed system, all region are connected using the P2P network and contain a copy of an entire blockchain. Here the architecture shows the intra (e-registry generated or transfer within a region) and inter (e-registry generated or transfer between two regions) P2P connection of the miners in different state regions. Each region has an individual/unique address in the network. The buyer/seller connects through the official portal of the “Land Registry Management Framework (LRMF).” By submitting the land transfer and land verification details and the buyer/seller requests the transaction. The SRO checks the details and verifies the seller/buyer’s information. Then SRO sends the verified document to the RO. The RO verifies the document and performs e-transaction. The transaction should be added to the pool of transactions. The miner creates a block consisting of transactions when a sufficient amount of transactions are available in the transaction pool. The new block created is broadcast by a leader miner to all nodes. Then the consensus is achieved on the transaction block. Finally, the agreed block is added to the blockchain.

3.2. Attributes of property transaction in block

Whenever any entity (Seller/Buyer) wants to transfer their land to other entities, they submit the details of the land by providing some information via the online portal. The details submitted by the users are stored in the form of a transaction. Fig. 3.3 shows the attributes that are stored in the form of a property transaction. A transaction contains the information related to the transaction, such as buyer and seller identities, property details, previous block hash, etc. The randomly selects a miner based on the miner’s trust value (discussed in 4.2). The selected miner is called a leader miner. A leader miner broadcasts the created transaction block to all nodes if submitted details are found valid. The consensus is then approved after voting the issued transaction block, and the block is added to the blockchain.

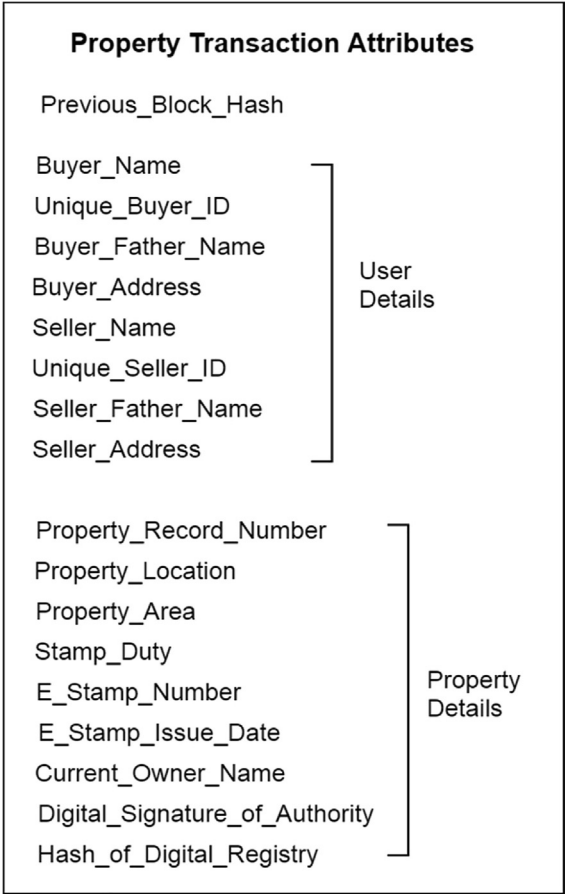


Fig. 3.3. Structure of Transactions in a Block.

3.3. Block creation for e-Registry generation transaction

The leader miner creates a block and adds verified transactions to store these transactions in the blockchain. All valid peers must

verify all transactions in the newly generated block to add this block to the blockchain. Once the miners validate all newly generated blocks, the consensus process is stated to achieve the final consensus to add this block into the blockchain. The next section proposes a novel approach to the process of trust value computation and consensus mechanism.

4. Leader selection process

The leader is responsible for maintaining the blockchain consistency by synchronizing the mining policies. In the proposed framework bootstrap server maintains the list of the active miner. Allocation of mining slots to miners is handled by the bootstrap server with the trust value table's (discussed in section 4.2 Table 1).

4.1. Proposed trust value based approach

The proposed e-registry transaction system comprises of two types of miners: one that is part of the G_LRO, and the second is APM. The node that is part of the G_LRO is termed here as Registry Office Miners (ROM). Here, ROM is classified into Highly Trusted ROM (HTROM) and Less Trusted ROM (LTROM) based on miners' trust values. The second type is the APM that invests in state-of-art infrastructure in these farms and provides their services to encash their investment.

4.2. Trust value computation

A fixed Trust Value (TV) (40 in our proposed approach) is assigned to every miner whenever they join the IPFS network for the first time. The TV table decides the registry office miners into two categories that are HTROM and LTROM. Each miner maintains the TV table. The leader miner uses this TV table to choose some reliable miners/nodes for the consensus process. The process of TV computation starts at every node simultaneously by getting the result of the final consensus. The TV of a node is computed based on the created block correctness and the verification response time. The TV computation algorithm is described below.

Algorithm 1: Trust value computation algorithm

```

1: Input: Final consensus result and votes from all
   consensus miners for block verification
2: Output: Updation of Trust Value (TV) table
3: Procedure: Computation algorithm ()
4:   Initial trust value (TV) = 40 //when join the network
   first time
5:   for each consensus miner (CM)
6:     if CM response = final consensus result // compute
       majority vote
7:       if TV is negative

```

(continued)

Algorithm 1: Trust value computation algorithm

```

8:     TV = 0
9:     else if TV is highest // proposed highest threshold
       is 120
10:       No change in TV
11:     else
12:       TV += 3 //if TV is between lower (0) and
       higher threshold (120)
13:     end if
14:     else
15:       TV -= 12 //CM response is invalid
16:     end if
17:     if response time (RT) <30 ms
18:       TV +=2
19:     end for
20: End procedure

```

Here miners, whose TV is less than 20 are considered less trusted miners/nodes. These miners will not get a chance to participate in the consensus process. The TV table is updated after each new block creation. TV is incremented or decremented based on the correct or incorrect verification of the new block and response time. Fig. 4.1 shows the flow diagram of trust value computation at each node after receiving the votes from the leader miner final results.

Each miner maintains a status table. The table classifies each node based on its trust value into three categories: HTROM, LTROM, and APM. The same is illustrated in Table 1. When the system starts, any arbitrary node assumes the miner role and broadcasts the newly created block to all other nodes along with its trust value. When an LTROM node trust value goes beyond 20%, its status is changed from the LTROM group to the HTROM group.

All of the other nodes update their status table by receiving the current TV of other miners. At some point of time, a node may be HTROM, but its trust value may decrease later, and now it finds its position in the LTROM list. Thus, the number of HTROM and LTROM nodes changes dynamically. The government-owned land registry office relies on HTROM and LTROM, although APM aims to increase operational efficiency.

4.3. Transaction verification process for the proposed system

Fig. 4.2 describes the procedure of adding a newly created block in the blockchain by miners in the network. When the buyer/seller wants to request a transaction, they must submit the details of their identities and the property. Then a block is created and broadcasted by a leader miner to all the nodes in the network. The consensus algorithm decides the selection of nodes for participating in the consensus. Furthermore, these nodes are called Consensus Miners (CM). Each CM performs a search operation in its respective blockchain until these find the block that contains trans-

Table 1
Miner Load Trust Value Status Table.

S. No.	HTROM		LTROM		APM	
	PEER ID	TV	PEER ID	TV	PEER ID	TV
1	A	40	C	19	E	62
2	B	54	D	17	F	48
3	G	44	P	15	R	60
4	M	54	S	20	K	50
5	X	60	T	18	W	46
6	–	–	–	–	–	–
7	–	–	–	–	–	–

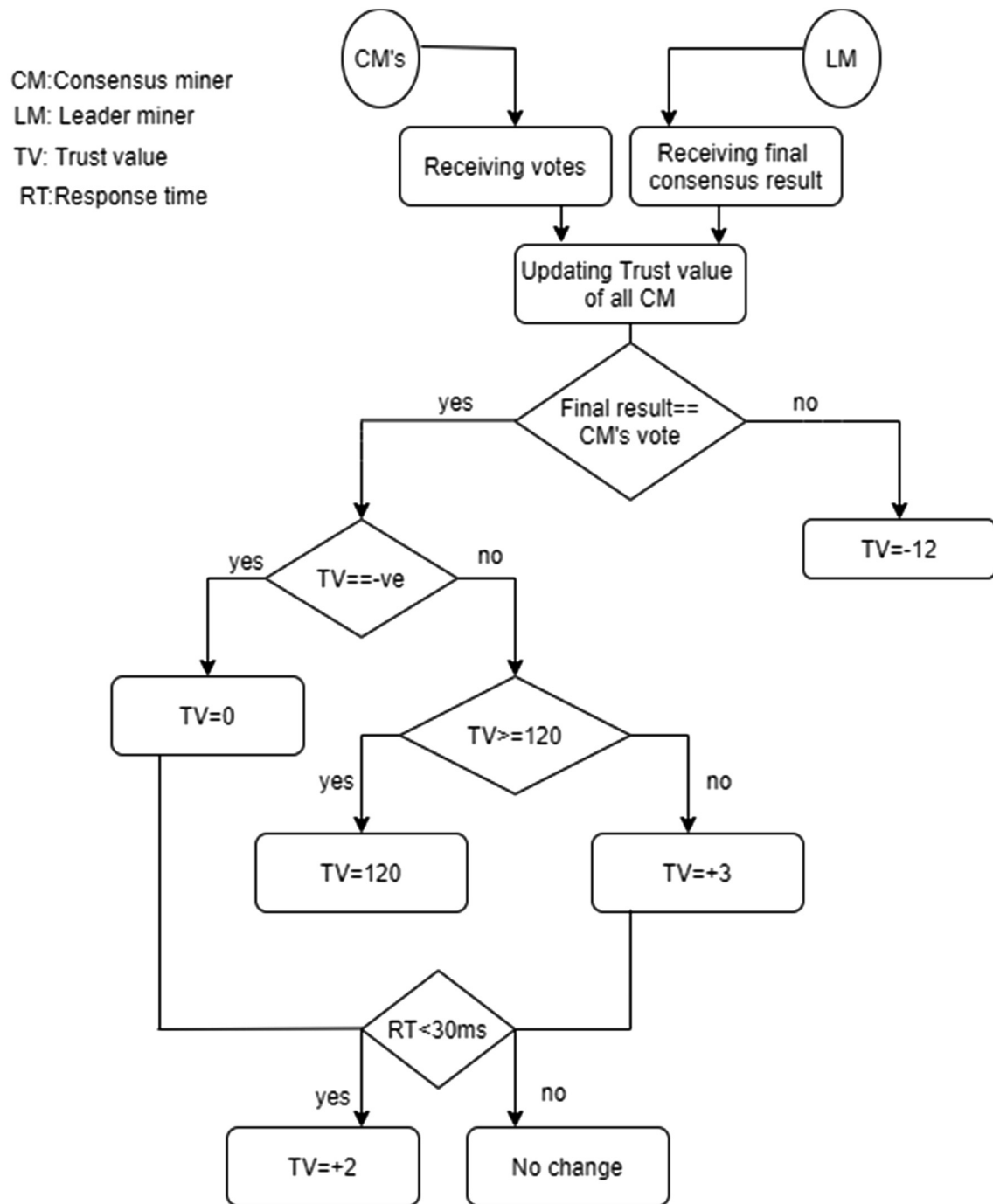


Fig. 4.1. Trust Value Computation.

actions where attributes match with the property details. Each CM response is either true (1) or false (0) based on the transaction details' validation. If the transaction is proved to be valid, then CM replies true, otherwise false. It sends the response to the leader miner. Each valid or invalid response of the miner is considered a vote. After getting the votes from all CM then the leader miner achieved a majority vote of 50% (HTROM) based on trust value, then the consensus is applied. If the final consensus result favours the block, then the block is added to the blockchain. Block is published and broadcasted to all nodes (consensus and remaining node) in the network. All other nodes add that block in their respective blockchain and update their trust value table.

4.4. Proposed Trusted nodes consensus algorithm

The existing consensus approach (PoW) requires lots of computation power and time to mine and add the block into the blockchain. We have proposed a trust value-based consensus algorithm for the land/property transaction processing system instead of the Proof-of-Work (PoW) approach. This proposed algorithm Trusted Nodes Consensus Algorithm (TNCA) minimizes the time required to add a block into the blockchain and requires less message exchange (discuss in section 5.2.1) than the existing PoW approach. Each node maintains the trust value table and updates it. The Trust Value (TV) table contains the node id (peer id) and their respective trust value.

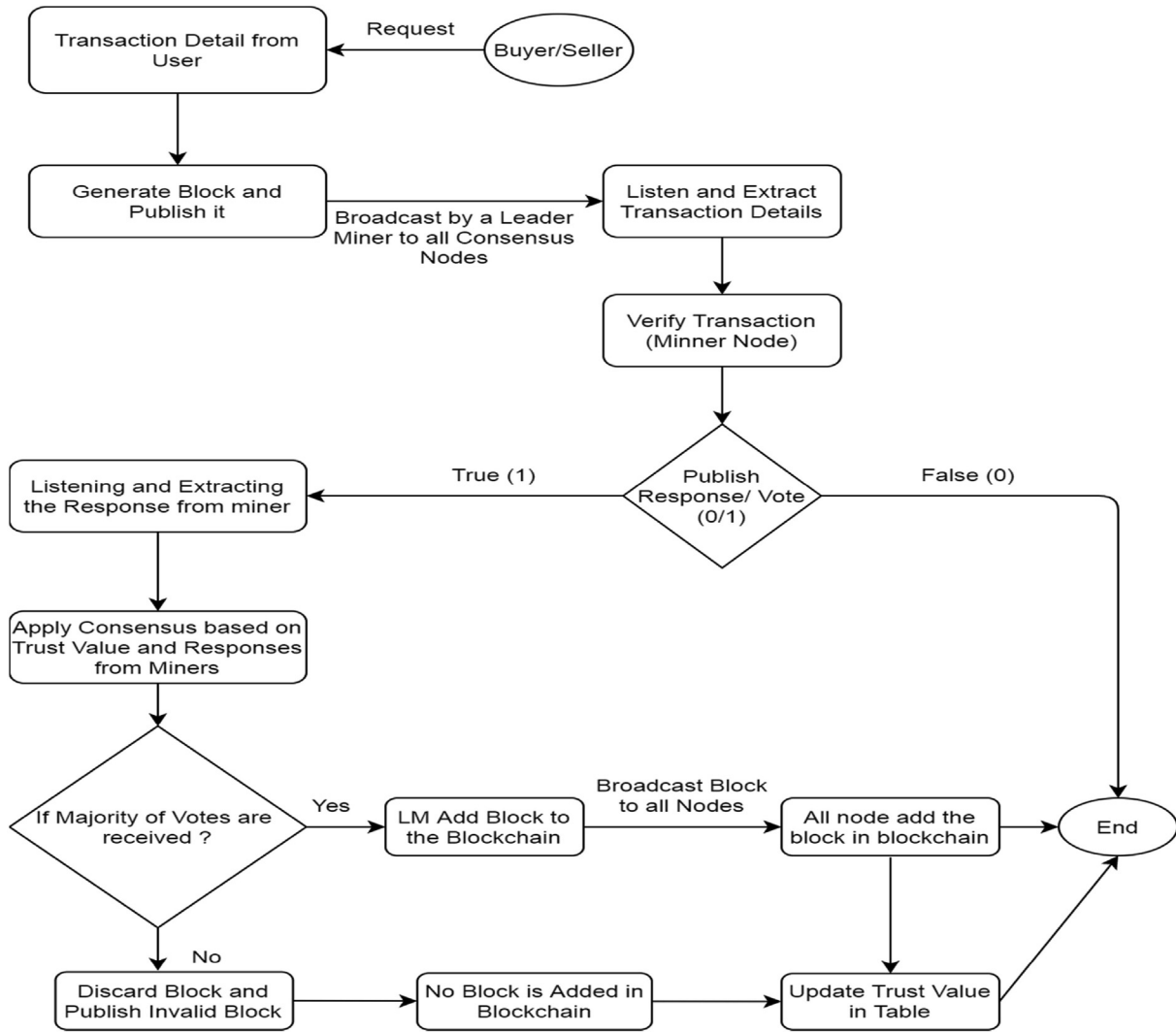


Fig. 4.2. Work Flow of the Proposed System.

Here, the leader miner randomly selects 50% from the HTROM and 50% of APM for the consensus process. CM broadcasts their votes with its peer id to all nodes (leader, consensus, and remaining) in the network. Therefore all nodes can update the trust values of each CM in the table after getting the final consensus result.

Algorithm 2: Trusted Nodes Consensus Algorithm (TNCA)

```

1: Input: Votes from Consensus Miners (CM) after verification of transaction
2: Output: Publish hash of added block otherwise discard block
3: Procedure: CONSENSUS APPROVAL ()
4:   sum = 0
5:   Broadcast block to all nodes by leader miner
6:   if this is the first block
7:     All nodes will be selected as consensus node
8:   else
9:     Select 50% of HTROM and 50% of APM randomly by leader miner
10:  end if
11:  //Continue receiving responses from consensus miners

```

(continued)

Algorithm 2: Trusted Nodes Consensus Algorithm (TNCA)

```

12:      //c = Number of miners selected for consensus
13:
14:      sum =  $\sum_{i=1}^c (\text{response} * \text{trust value})$ 
15:
16:      if sum > 0 & majority votes is received in favour of block addition
17:        Add the block and broadcast the block to all the nodes.
18:      else
19:        No block added and publish invalid response
20:      end if
21:  End procedure

```

Based on the calculation and the majority of votes, the final consensus result is achieved. If the block is proved to be valid, then-leader miner adds the block into its existing blockchain and broadcasts the block to all the nodes. Therefore other nodes also add that block into their respective blockchain. Based on the trust value, the leader miner's responsibility is to select the miners to verify the newly cre-

ated block. Each miner maintains the status table based on trust value.

5. Performance evaluation

Performance evaluation of the proposed approach - TNCA is required to determine its scalability and robustness compared to the existing proof PoW approach.

5.1. Number of block vs mining time

The number of miners is kept fixed, and the number of blocks in the blockchain is varied. The miners need to traverse the blockchain to fetch information about the property and users. Miners return the validity status of the transaction. Fig. 5.1 shows the comparative analysis between the number of blocks and their corresponding mining time.

5.2. Performance comparison with existing approaches

The performance is measured by changing the number of miners. The number of blocks traversed by each miner to fetch the buyer and seller information is kept fixed. Some blocks are already added into the blockchain, and a new transaction is made intentionally so that each miner traverses a fixed number of blocks. The number of miners is incremented subsequently, and the execution time in seconds or the number of messages exchange is noted. The performance result of the proposed consensus approach - TNCA is compared with PoW on the basis of the following measures:

1. Number of Message Exchange,
2. Time (in milliseconds).

5.2.1. Comparison with PoW

The performance of the proposed approach - TNCA is measured in terms of the number of messages required to achieve the consensus. Table 2 shows the total number of messages required for the final consensus by the proposed approach - TNCA and PoW. Suppose there is a network of 'N' nodes. In the PoW approach, all 'N' nodes participate in the consensus, and after verification, each broadcast its vote to all 'N-1' nodes. Therefore PoW requires a total $N*(N-1)$ message exchange. Whereas the TNCA requires fewer message exchanges due to fewer 'M' nodes selected for taking part in the consensus process. This approach causes the exchange of $M*(N-1)$ messages. Let N_1 and N_3 be the total number of miners in the group of HTROM, and APM. Let X_1 and X_3 be the group of selected nodes for the consensus process from N_1 , and N_3 . Where X_1 , and X_3 are defined as $(N_1*50)/100$, and $(N_3*50)/100$ respectively. The

number of miners in group X_2 is considered 0 because, based on the proposed approach - TNCA, nodes from less trusted categories do not get a chance to participate in the consensus process. Hence, the Total Number of Messages exchange (TNME) required for the proposed approach - TNCA are:

$$TNME = (X_1 + X_2 + X_3) * (N - 1)$$

Table 2 illustrate that the number of message exchanges required by the TNCA is 60% lesser compared to the existing PoW approach.

The right side of the Fig. 5.2 represents the message flow in Proof of Work (PoW), where all miners send responses to each other. The figure's left side shows the message flow in the proposed approach - TNCA after the verification of block where one node is LM, and few nodes are selected as CM based on trust value. Only CMs are sending responses to the LM.

LM sends two types of messages: The LM sends the first message to the selected CMs based on trust value. After the transaction is verified and authenticated, then the CMs send the vote LM. Second, the LM receives the majority of votes from the CMs. Then the LM broadcasts the message to all nodes in the network.

The Initial messages from the leader to all nodes are not included in the figure as those numbers of messages are the same in both approaches. It is visible that the number of message exchanges required in the TNCA is very less compared to PoW.

Fig. 5.3 represents the graphical view of the comparison of the PoW and our TNCA. The proposed approach - TNCA requires an average of only 58.94% of message exchange per consensus process compared to the traditional PoW approach. The reason behind the better performance in the TNCA is selecting fewer nodes based on their previous performance for the consensus process among all nodes in the network.

5.2.2. Performance evaluation comparison with PoW, PoS, DPoS and TNCA

The proposed approach - TNCA execution time is further compared with the traditional approaches, such as PoW, PoS, and DPoS. Table 3 compares the PoW, PoS, and DPoS with the TNCA on the execution time of the experiments. Table 3 shows the execution time of the PoW, PoS, DPoS, and TNCA. The number of miners is incremented from 10 to 350, and the execution time is noted. The tabulation is for the results obtained from different simulations for computing execution time.

The graphical representation in Fig. 5.4 shows the number of nodes varying from 10 to 350. Compared to the PoW, PoS, DPoS, and proposed approach - TNCA, the TNCA performs better in term of execution time. The result deduces that the TNCA reduces the execution time by 67.59%, 61.65%, and 47.99% in comparison to PoW, PoS, and DPoS approaches.

5.2.3. Comparison with trust based consensus approach

Singh and Vardhan (Singh et al., 2019) in 2020 proposed a trust-based consensus approach based on message exchanges. The number of transaction requests decides the miner into two category namely, HTROM and LTROM. 50% of lightly loaded miners and 50% of private miners use the method as a load-based consensus miner. This approach focuses more on CPU utilization. Moreover, the proposed approach - TNCA focuses more on reliability and trust factor. Both approaches require almost the same number of message exchanges for achieving consensus because both are taking on an average of 50% of miners as consensus miners. The difference can be measured in terms of time as the load-based approach maintains the load status table. It consume time by getting and updating the miner's load status. Table 4 shows the comparative analysis of the TNCA and load-based approach in seconds to achieve the complete process. The number of miners is incre-

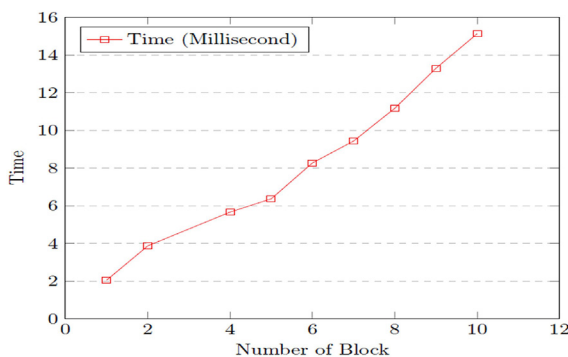


Fig. 5.1. Number of Block Vs Mining Time.

Table 2
Comparison of the TNCA and PoW approach in terms number of message exchange.

S. No.	N	N1	X1	N2	X2	N3	X3	TNME (PoW)	TNME (TNCA)	Message reduction (%)
1	10	5	3	2	0	3	2	90	45	50
2	20	9	5	4	0	7	4	380	171	55
3	30	14	7	6	0	10	5	870	348	60
4	50	22	11	11	0	17	9	2450	980	60
5	100	45	23	22	0	33	17	9900	3960	60
6	150	67	34	33	0	50	25	22,350	8791	60.6
7	200	90	45	44	0	66	33	39,800	15,522	61
8	250	111	56	55	0	84	42	62,250	24,402	60.8
9	300	134	67	66	0	100	50	89,700	34,983	61
10	350	156	78	77	0	117	59	122,150	47,813	61

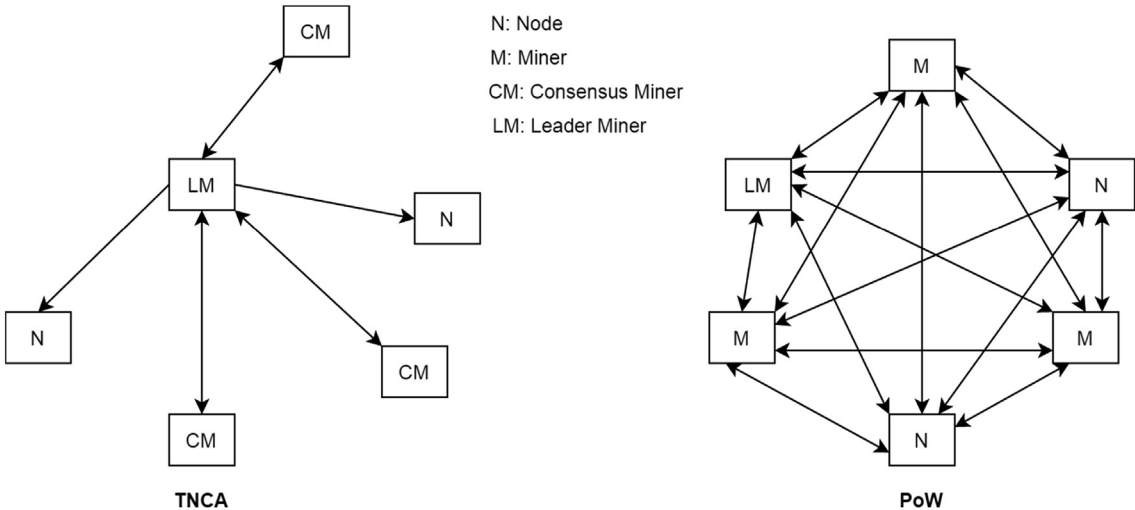


Fig. 5.2. Message Flow for Consensus Votes.

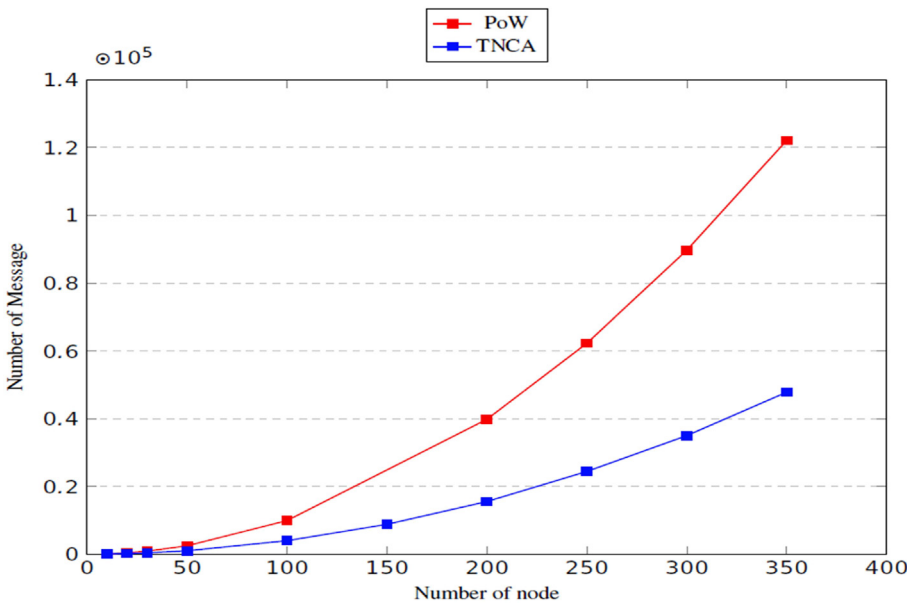


Fig. 5.3. Analysis of PoW and TNCA in Terms of Messages Exchange.

mented from 10 to 1000, and the execution time is noted. The tabulation is for the results obtained from different simulations for computing execution time.

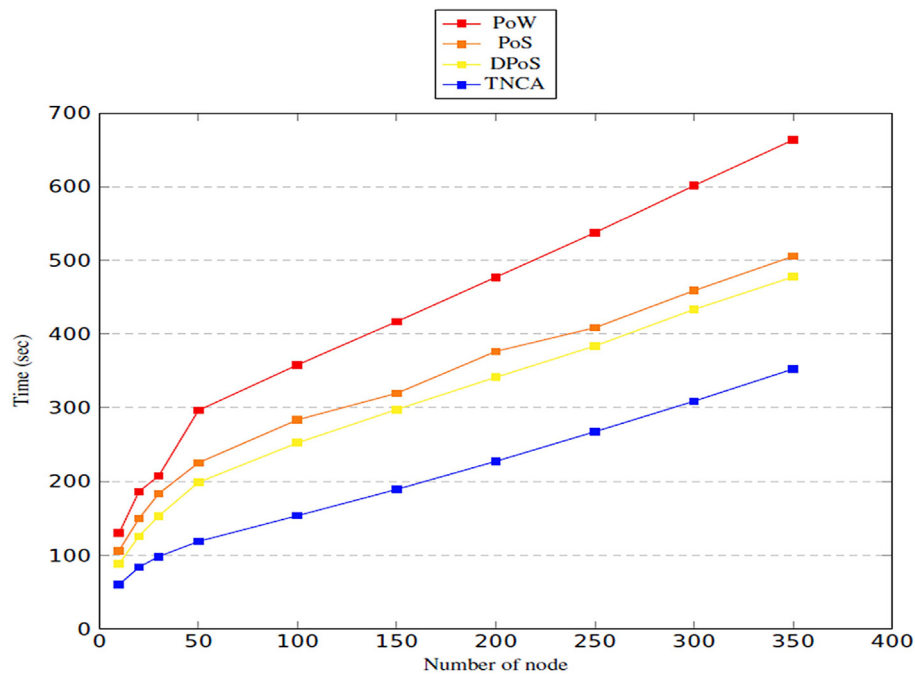
Fig. 5.5 shows the graphical representation of the comparative assessment of the proposed approach - TNCA and load trust-

based approach. The number of nodes ranges from 100 to 1000. The results deduce that the TNCA reduces the execution time by 26.44% compared to the load and trust based approach. Compared to the state-of-the-art algorithms, the TNCA performs better.

Table 3

Time analysis between PoW, PoS, DPoS and TNCA in seconds.

S. No.	No. of Node (N)	PoW	PoS	DPoS	TNCA
1	10	129.8	106.2	88.3	60.1
2	20	186.2	149.6	125.6	83.5
3	30	207.6	183.4	153.2	98.3
4	50	296.4	225.3	198.7	118.6
5	100	358.2	283.7	252.4	153.6
6	150	416.6	319.6	297.6	189.2
7	200	476.9	376.3	341.3	227.3
8	250	537.3	408.4	383.6	267.5
9	300	601.6	458.7	433.2	308.9
10	350	663.7	505.3	477.4	352.6

**Fig. 5.4.** Comparison of the Number of Blocks Generated by the PoW, PoS, DPoS and TNCA.**Table 4**

Analysis of TNCA and existing trust based approach in seconds.

Sr. No	No. of Node (N)	Load and Trust based Approach (Singh et al., 2019)	TNCA	Execution Time Reduction (%)
1	100	33	24	27.3
2	200	32	23	28.1
3	300	31	22	29
4	400	32.5	22.4	31.1
5	500	32	21.4	33.1
6	600	32.5	23.3	28.3
7	700	33.4	23	31.1
8	800	31.4	24	23.5
9	900	32.2	21.6	32.9
10	1000	32	22	31.2

6. Conclusion

This research provides a scalable and novel mechanism for transacting the property/land from one entity to another. Blockchain is used as the underlying technology in this research and can be adopted by present-day land registry sectors with P2P integration. Features of blockchain can provide transparent, secure, fast processing, and decentralized system. This article proposes an efficient consensus algorithm based on trust value. On an average, 50% of nodes take part in the proposed trust value-based con-

sensus mechanism. It makes the system scalable as the time taken for the consensus process is approximately 58.94% lesser than the traditional PoW approach, and 26.44% lesser time is taken as compared to the existing load-balanced method. Due to the message exchange, communication overhead for the complete process is also reduced by 60% on an average. So, the performance of the proposed trusted node consensus algorithm framework efficient than PoW. Hence, the blockchain technology-based property transaction system is highly scalable and secure for intrusive and fraudulent activities and can be deployed in real-time applications.

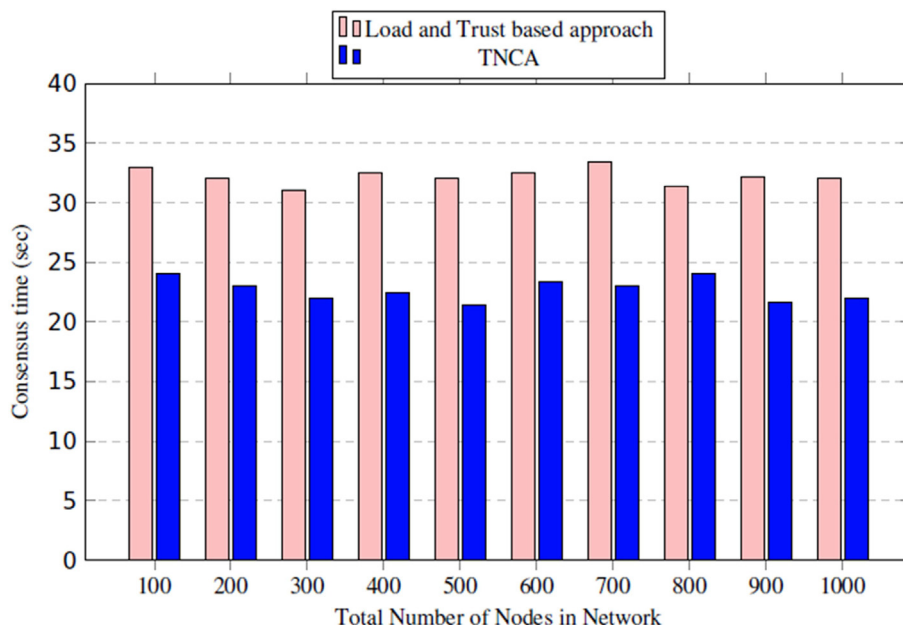


Fig. 5.5. Analysis of Load and Trust Approach and TNCA in Seconds.

There are also a few disadvantages to the blockchain, such as there is no generic and uniform blockchain architecture. The analysis should be performed in these directions. The load on the blockchain scheme will steadily increase and, due to population growth, transactions will increase. Storing these huge blockchain and performing efficient search operation requires efficient search algorithms.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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